Block Bishop

Input file:	`standard input`
Output file:	`standard output`
Time limit:	1 second
Memory limit:	256 megabytes

Evirir the dragon is playing chess on a board with 2 rows and N columns. The rows are numbered 1 and 2 from bottom to top, and the columns are numbered $1, 2, \ldots, N$ from left to right. Cell (r, c) denotes the cell in row r and column c.

A bishop stands on cell (X, Y). As in chess, it moves diagonally. Since the board has only 2 rows, each move takes it to a diagonally adjacent cell, that is, from (r, c) to $(3 - r, c - 1)$ or $(3 - r, c + 1)$. It cannot leave the board or move onto a blocked cell.

Before the bishop makes any move, Evirir must **block exactly one cell** other than (X, Y). After that, the bishop may make any number of moves (possibly zero). A cell is *reachable* if it is possible to move the bishop to it after any number of moves. The starting cell (X, Y) always counts as reachable.

If Evirir chooses the blocked cell optimally, what is the minimum possible number of reachable cells?

Input

Each test contains multiple test cases. The first line contains the number of test cases T ($1 \le T \le 1000$). The description of the test cases follows.

The first and only line of each test case contains three space-separated integers N, X, and Y — the number of columns, and the row and column of the bishop's starting cell, respectively.

Output

For each test case, output a line containing a single integer — the minimum possible number of reachable cells.

Scoring

For all tests, the input will satisfy the following constraints:

- $1 \le T \le 1000$
- $1 \le N \le 10^{18}$
- $1 \le X \le 2$
- $1 \le Y \le N$

Output Task

Subtask	Output file(s)	Points	Additional constraints
0	`out_sample_01.txt`	0	Samples.
1	`out_01.txt`	10	$N \le 5$, $T = 3$
2	`out_02.txt`	10	$N \le 10$, $T \le 10$
3	`out_03.txt`	10	$N \le 100$, $T \le 10$
4	`out_04.txt`	10	$N \le 100$
5	`out_05.txt`	10	$N \le 2 \times 10^5$, $T \le 10$
6	`out_06.txt`	10	$N \le 2 \times 10^5$
7	`out_07.txt`	10	$N \le 10^9$, $Y \le \frac{N+1}{2}$, $T \le 10$ (the bishop starts in the left half)
8	`out_08.txt`	10	$N \le 10^9$, $T \le 10$
9	`out_09.txt`	10	$N \le 10^9$
10	`out_10.txt`	10	—

Example

standard input	standard output
3 5 2 2 1 1 1 6 1 3	2 1 3

Note

Sample 1

This sample is valid for output subtasks 2–10.

In the first test case, $N = 5$ and the bishop is on $(2, 2)$. Evirir blocks $(1, 3)$. The bishop can move to $(1, 1)$ and nowhere else, so 2 cells are reachable.

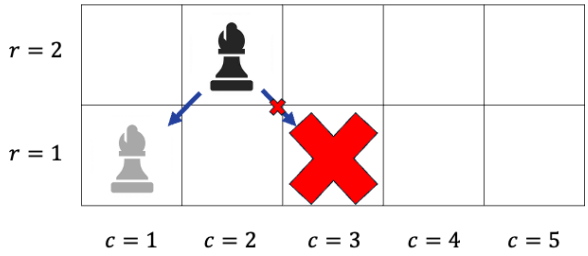


Illustration of test case 1

In the second test case, $N = 1$. Evirir must block $(2, 1)$, the only other cell. The bishop has no legal move, so only 1 cell is reachable.

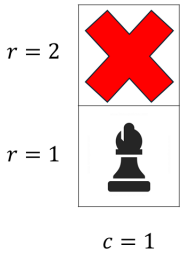


Illustration of test case 2

In the third test case, $N = 6$ and the bishop is on $(1, 3)$. Evirir blocks $(2, 4)$. The bishop can reach $(2, 2)$ and $(1, 1)$, so 3 cells are reachable. Blocking $(2, 2)$ instead would let the bishop also reach $(2, 4), (1, 5), (2, 6)$, so 4 cells would be reachable, which is worse.

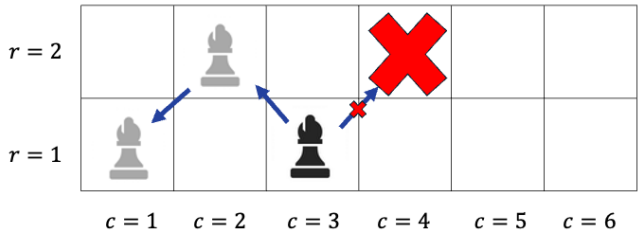


Illustration of test case 3